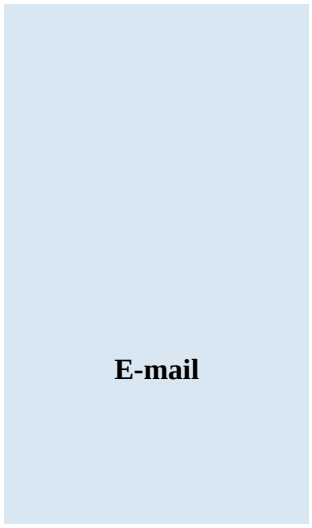


一种博弈驱动的多智能体语义共识方法及影视推荐系统

E-mail

2026 10

0 缩略语和关键术语定义

术语	定义
AEIC	Assumption Evidence Inference Conclusion
	Agent
	Agent AEIC /
/	
	Agent
	Agent P2P
	Agent

1 相关技术背景及最接近的现有实现方案

1.1

Agent

Agent Agent “ ” “ ”

Agent Top-K Agent

1.2

1.2.1 现有实现方式

Agent

1.2.2 现有方案的不足

1 Agent

2

3

4

1.3 Agent

1.3.1 现有实现方式

Agent

1.3.2 现有方案的不足

1 Agent

2

3 Agent Top-K

4

1.4

1.4.1 现有实现方式

1.4.2 现有方案的不足

1 Top-K

2

3

4

1.5

现有方案	本发明的区别特征	产生的技术效果
Agent	Token	

□□□□	□□□□□□□□	□□□□□□□
	A/E/I/C	
	Agent	

2 □□□□□□□□□□□□

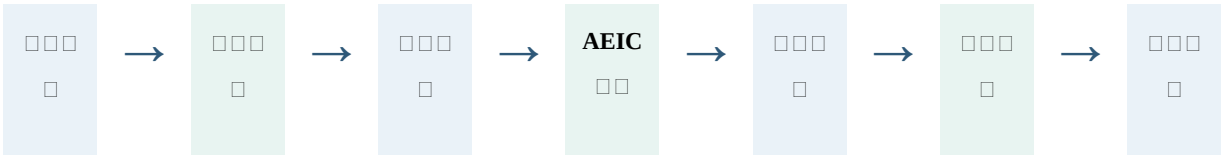
2.1

Agent Agent

Agent

Top-K

2.2



1

1 “ ” Trace Agent

Token “ ”

2.2.1 □□□□□□□□□□□□

q Agent N={1,…,n} t={1,…,T} S_t

Agent Agent

$$S\_t = (q,\ m,\ t,\ \rho(t-1),\ P\_t,\ B\_t,\ H\_t)$$

ρ(t−1) t P_t B_t H_t Agent i

view_i(S_t) Agent Agent

τ_c τ_a τ_r τ_c<τ_a<τ_r Agent

Agent

□□	□□
q N T	Agent

符号	含义
S_t H_t	t
m=(g,c,P,B)	
a_i R_i	Agent i
r_i s_i	AEIC
ρ_i ω_i	Agent
C_frozen z_ij	Agent i j

2.2.2 步骤 S1：构造并公开合同

m g c P B Agent

$$m = (g, c, P, B)$$

Agent

Token Agent

Token

JSON g c P

Agent B Token

合同分量	内容	执行约束
g		
c	Low AEIC Verify Imitate	
P	Agent× ×	Agent
B	Token	

2.2.3 步骤 S2：Agent 独立选择行动

Agent A Low AEIC Verify Imitate

行动	真实执行路径	可记录资源
Low	Token	/ Token
AEIC		Token

行动	真实执行路 □	可记录资源
Verify	AEIC	
Imitate		/

Agent action reason_code expected_credit expected_cost reason_code expected_credit expected_cost Agent action Low

$$a\_i(t) \in \{Low, AEIC, Verify, Imitate\}$$

2.2.4 步骤 S3：按行动路由实际资源

Agent Verify Low Token

DeepSeek API Agent “ ” “ ”

action executor Low AEIC Verify AEIC Imitate P Agent action_trace

$$R\_i = route(a\_i, P\_i, B\_i)$$

R_i Agent i a_i R_i Verify Imitate B_i Agent

2.2.5 步骤 S4：生成 AEIC 结构化语义报告

AEIC Assumption Evidence Inference Conclusion / /

字段	主要内容	可计算信号
A		field_A
E		field_E novel_evidence verified_evidence
I	/	field_I consistency useful_correction
C		field_C unsupported_flip

AEIC “ ” “ ”

Evidence evidence_id claim source_type source_ref status span source_type source_ref status span verified_evidence

Conclusion / / support oppose abstain preference_score risk_flags Trace

2.2.6 步骤 S5：提取可观察语义信号

s_i

A E I C Agent

$$completeAEIC\_i = 1[A\_i \neq \varnothing \wedge E\_i \neq \varnothing \wedge I\_i \neq \varnothing \wedge C\_i \in Y]$$

$$novelEvidence\_i = |tokens(E\_i) \setminus tokens(E\_visible, i)| / max(1, |tokens(E\_i)|)$$

useful_correction Agent unsupported_flip

$$unsupportedFlip\_i(t) = 1[C\_i(t) \neq C\_i(t-1) \wedge \Delta E\_i(t) = 0 \wedge verify\_i(t) = 0]$$

verified_evidence

unknown

2.2.7 步骤 S6：计算信用、成本和已实现效用

AEIC

$$credit\_i = \Sigma\_k \; w\_k \cdot s\_k(r\_i, h\_i, y)$$

$$utility\_i = credit\_i - cost\_i(action\_i)$$

r_i Agent i h_i Agent y

AEIC

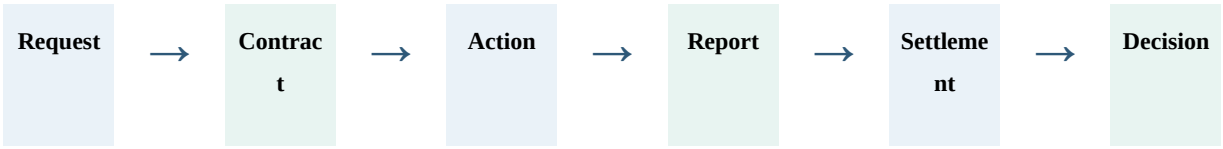
$$positive\_i = w\_c \cdot complete\_i + w\_n \cdot novel\_i + w\_v \cdot verified\_i + w\_u \cdot correction\_i$$

$$credit\_i = clip(positive\_i - w\_f \cdot flip\_i, L, U)$$

y outcome

2.2.8 步骤 S7：批量更新信誉

$$reputation\_i(t) = decay \cdot reputation\_i(t-1) + utility\_i(t)$$



j reputation_before Agent u_ij

reputation_after

$$\rho\_i(after) = decay \cdot \rho\_i(before) + mean\_j(u\_ij)$$

agent_id

2.2.9 步骤 S8：形成可审计集体决策

Agent

y Agent ω_i Agent

support oppose

$$\omega\_i = clip(conf\_i \cdot (1 + \rho\_i / K) \cdot max(0, utility\_i),\ \omega\_min,\ \omega\_max)$$

$$consensus(y) = \Sigma\_i\ \omega\_i \cdot 1[C\_i = y] - \lambda \Sigma\_i\ \omega\_i \cdot 1[C\_i\ opposes\ y]$$

Agent

“ ”

2.2.10 数据结构和审计 Trace

2

request_id Trace AEIC

记录对 □	必需字段示例
	request_id
	contract_version g c P B committed_at
	agent_id action visible_state selected_at
	A/E/I/C evidence_refs raw_response tokens latency
	signals credit_breakdown cost utility reputation_before/after
	normalized_result support/oppose residual_dissent rank score_components

Trace event_id request_id event_type actor timestamp payload_hash prev_hash

Trace

$$hash\_e = H(request\_id \;||\; event\_type \;||\; payload\_hash \;||\; prev\_hash\_e)$$

Agent× ×

complete incomplete

2.2.11 预算约束和执行终止条件

Token Token Agent

$$resource\_i = \eta\_in \cdot tokenIn\_i + \eta\_out \cdot tokenOut\_i + \eta\_call \cdot calls\_i + \eta\_v \cdot verify\_i$$

$$\Sigma\_i \; resource\_i \le B\_total \quad \wedge \quad resource\_i \le B\_i$$

AEIC

Trace

2.3

Agent Trace

Agent

模块	输入	输出	关键约束
	Top-K		
			ID
			Agent
Agent		AEIC	
		/	
Trace			

2.3.1 请求解析与风险预算规划

/

Top-K

Agent

query required_genres preferred_genres excluded_genres min_year max_year top_k “ ” excluded_genres “ ” preferred_genres

Agent

$$risk = clip(b\_0 + b\_h \cdot n\_hard + b\_n \cdot n\_negative + b\_q \cdot complexity + b\_k \cdot topK, 0, 1)$$

2.3.2 真实目录召回与冻结候选集

18,430 MovieLens

Top-K Agent candidate_id

Top-K

ID

movie_id title year genres retrieval_score catalog_evidence

$$C\_frozen = topL(\{x \in Catalog \mid hardPass(x,q)=1\}, retrievalScore)$$

L top_k candidate_id L Agent C_frozen

2.3.3 多角色 Agent 对统一候选集进行 AEIC 评审

Agent Agent Agent Agent Agent AEIC

Agent

j z_j z_ij=1 Agent i z_ij=0 N_req j coverage_incomplete

Agent

$$coverage(j) = \Sigma\_\{i \in N\_req\} \; z\_ij \; / \; |N\_req|$$

2.3.4 硬约束门控与候选级共识排序

Agent

“ Top-K ”

$$score\_j = \alpha \cdot retrieval\_j + \beta \cdot agentUtility\_j + \gamma \cdot evidence\_j - \mu \cdot opposition\_j$$

opposition_j Top-K

hardPass_j

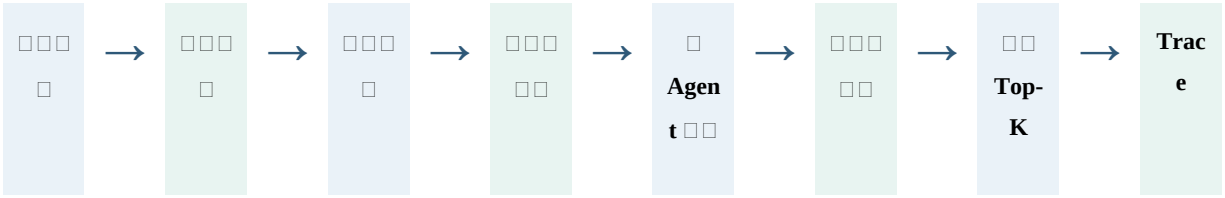
Agent

$$hardPass\_j = yearPass\_j \wedge requiredGenrePass\_j \wedge excludedGenrePass\_j$$

$$TopK = argsort\_K(\{score\_j \mid j \in C\_frozen \wedge hardPass\_j = 1 \wedge coverage(j) = 1\})$$

K score_components

Agent



3

2.3.5 □□□□□□□□□□□□□□□□

“ ” Agent AEIC Trace

Token A/E/I/C

request_id Trace

2.3.6 □□□□□□□□□□□□

HTTP

request_id

Agent

mechanism_status degraded “ Agent ”

□□	□□	**Trace □□**
	Low	invalid_action fallback
AEIC		schema_repair raw_hash
Agent		timeout retry coverage
	Top-K	verification_unavailable
		degraded missing_agents
	request_id	client_disconnected

2.4

2.4.1 通用多智能体判断实施例

AEIC Verify Low Imitate

Agent Agent

A E I C P

Agent Agent

Agent Verify unsupported_flip

useful_correction verified_evidence

阶段	控制器动作	**Agent** 或工具输出	审计重点
	g c P B		
		action	
		Token	
		A/E/I/C	

2.4.2 多约束影视请求实施例

“ 2010 ” Top-K 3 min_year=2010 required_genres={Science Fiction, Drama} excluded_genres={Action, Disaster} “ ”

18,430 L movie_id

AEIC

Science Fiction Drama Action hardPass Top-K evidence

“ ” A/E/I/C

2.4.3 关键算法流程

1 q request_id

2 m committed

3 Agent view_i(S_t) a_i route(a_i)

4 AEIC s_i

5 g credit_i c(a_i) utility_i

6

7 reputation_after

8 complete incomplete

2.5

Agent StrategyQA MMLU-Hard TruthfulQA 150 450 5 Agent 3 Seed

比较条件	高资源行动率	第三轮多数决策准确率
No Mechanism	0.46%	68.00%
AEIC Static AEIC	56.93%	82.22%
	+56.47	+14.22

Exact McNemar Holm $p=1.23\times10^{-11}$ “ ”

30 150 Top-5 Agent

StrategyQA 73.33% 77.33% MMLU-Hard 71.33% 91.33% TruthfulQA 59.33% 78.00%

3 5 Agent AEIC

2.6

1 Agent

2 AEIC

3 Verify AEIC Token

4 –

5

6

7 Trace

2.7

1 Agent

2

3

4 Agent

5 Agent

6 Agent Trace Agent

7 AEIC JSON

8

9

3 本发明的关键技术点和拟保护点

1.

2. – Agent Token

3. AEIC

4. –

5.

6. Agent

7. Agent Top-K

8. request_id AEIC

9.

10.

Top-K

11. AEIC

Trace

“ ”“ ”“AEIC ”“

Trace” Agent

4 附图说明及技术材料索引

4.1

1 1 AEIC Trace

2 2 request_id

3 3 Top-K

4 4 Agent

4.2

材料	位置/□ 明
	docs/papers/ experiments/results/ AEIC
	system/game_consensus_recsys/ Agent HTTP Trace
	experiments/results/movie_acceptance/ acceptance_results.jsonl 30
	docs/patent/source/ .docx
	docs/patent/source/ .docx

4.3

Agent

拟保护技术特征	主要说□□支撑	可验证载体
	2.2.2 2.2.3 1	request_id
	2.2.4 2.2.5 1	Prompt Token
AEIC	2.2.6 2.2.7 2	
–	2.2.8 2.2.10	
	2.3.2 2.3.5 3 4	ID
Trace	2.2.10 2.3.6 2	